Materia

ANALYSE • VISUALISE • UNDERSTAND

Quick tutorial · version 1.2

From raw instrument files to dissertation-ready figures

DSC · tensile / compression · cyclic superelastic tests
SEM / LOM micrographs · elastocaloric analysis

1 Import
CSV / TXT from DSC, tensile, cyclic

→

2 Plot
overlay, style, customise

→

3 Interact
cursor, peaks, integrate, fit

→

4 Analyse
As/Af/Ms/Mf, E, UTS, fatigue

→

5 Save & export
session, PNG 600 dpi, SVG, PDF, CSV

Micrographs tab (separate workflow): open → calibrate → adjust → threshold → particles / grain size / phase fraction → export

Install once

Install **Python 3.10+** (Windows: tick Add Python to PATH). Unzip Materia, then double-click **Start Materia (Windows).bat** or **Start Materia (Mac).command**. The first start shows the Materia loading screen while it sets everything up, then puts a **Materia** shortcut on your Desktop.

Try it in 30 seconds

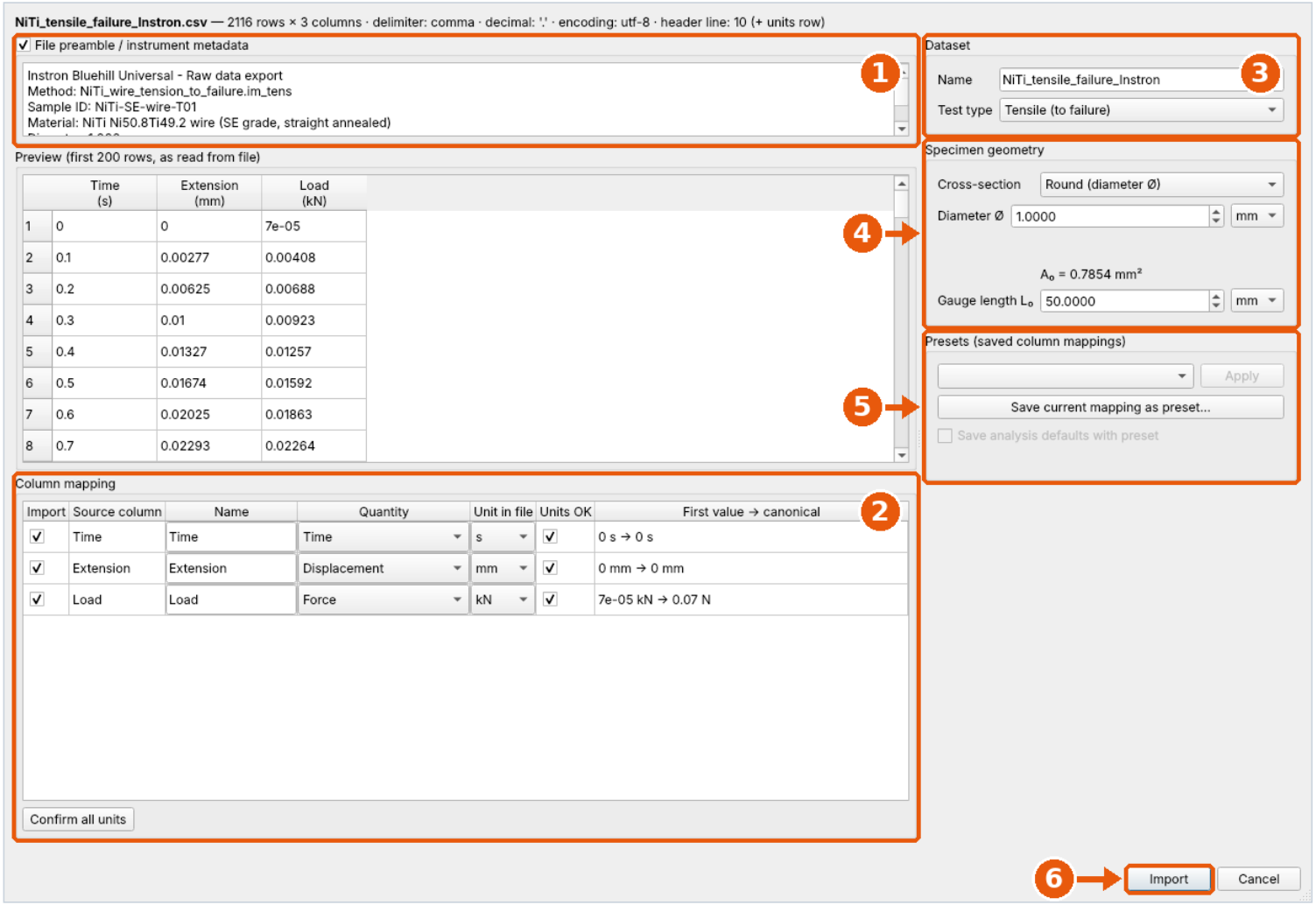
File ▸ Load sample NiTi data loads a synthetic DSC scan, tensile-to-failure test, compression test, 50-cycle superelastic test and an SEM grain image, so you can try every feature before using real data.

One rule to remember

Every value is stored in consistent units: **N, mm, mm², MPa, %, °C, s, mW, W/g**. You tell Materia the unit each column was recorded in once (and save it as a preset); everything downstream is then already in the right unit.

Shortcuts: Ctrl+S save session · Ctrl+O import data · Ctrl+Shift+O micrograph · Ctrl+E export plot · Ctrl+, preferences · F1 this tutorial · drag & drop files or .lstudio sessions

Importing a CSV / TXT file



1 Instrument header
Lines above the data (sample ID, mass, heating rate, Ø, gauge length) are read automatically and used to pre-fill the forms.

2 Column mapping
One row per column: tick to import, rename, choose the **Quantity** and the **Unit in file**. Guessed units are **amber**: tick Units OK or press Confirm all units. The last column previews the conversion (7e-05 kN → 0.07 N).

3 Test type
DSC, tensile, compression, cyclic or generic. It decides the default axes and which analysis tab the dataset appears in.

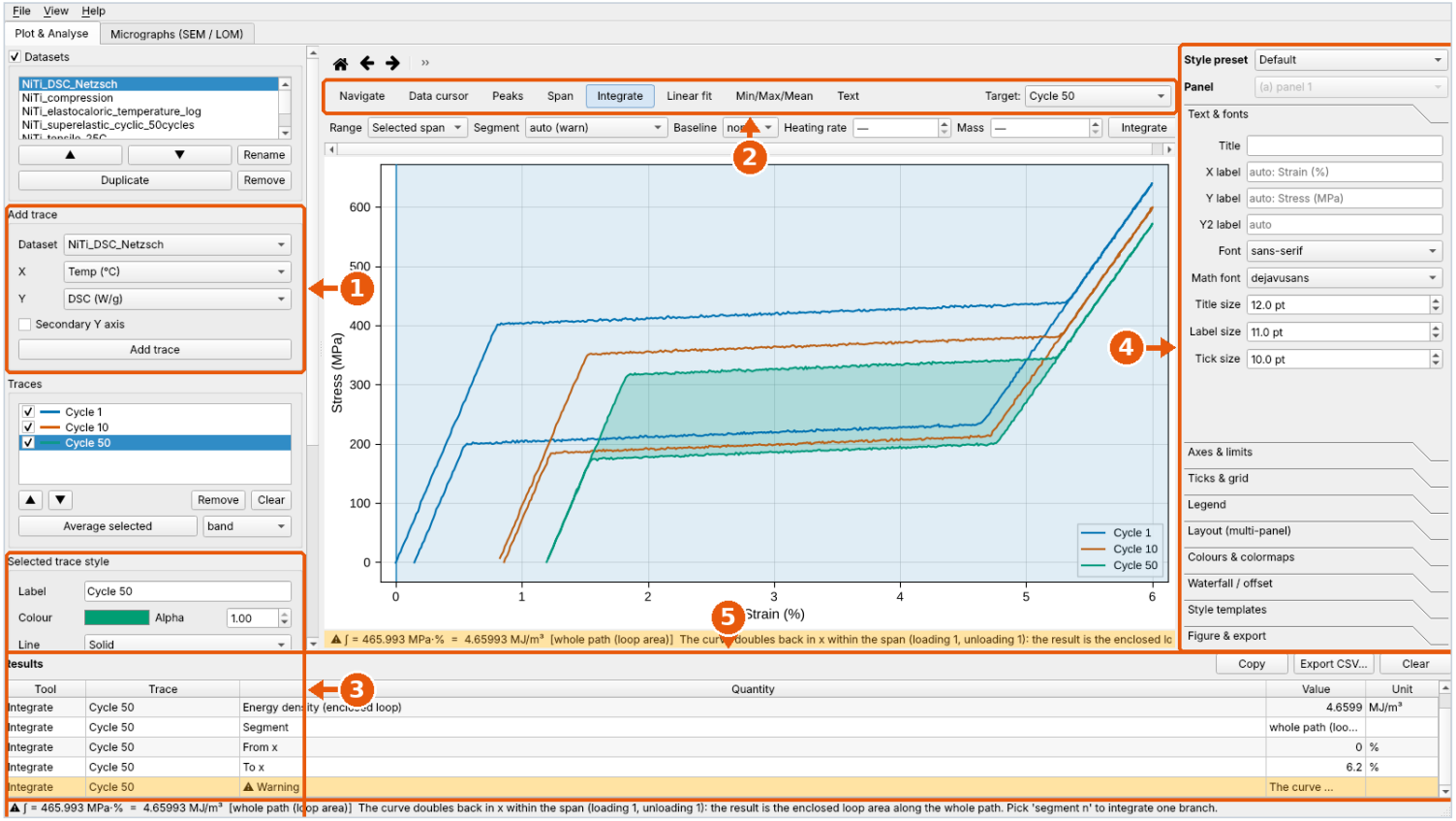
4 Specimen / sample
Mechanical: area, round Ø or width × thickness, plus gauge length L_0. DSC: mass and heating rate. If no rate is in the file the field turns amber; confirm it, or a warning follows the data.

5 Presets
Save current mapping as preset… once per machine (optionally with your analysis settings); next time pick it and click Apply.

6 Import
The dataset is added and plotted straight away. Several files? Tick Apply to all remaining files.

> **Tip:** CSV, TXT and **Excel (.xlsx/.xls/.ods)** with a sheet picker. Spelled-out units (minutes, kilonewtons) are understood, and MJ is never confused with mJ.

Plotting and customising graphs



1 Add a trace
Pick dataset, X and Y column (units shown), then **Add trace**. Overlay as many curves from as many files as you like; tick Secondary Y axis for a right-hand axis.

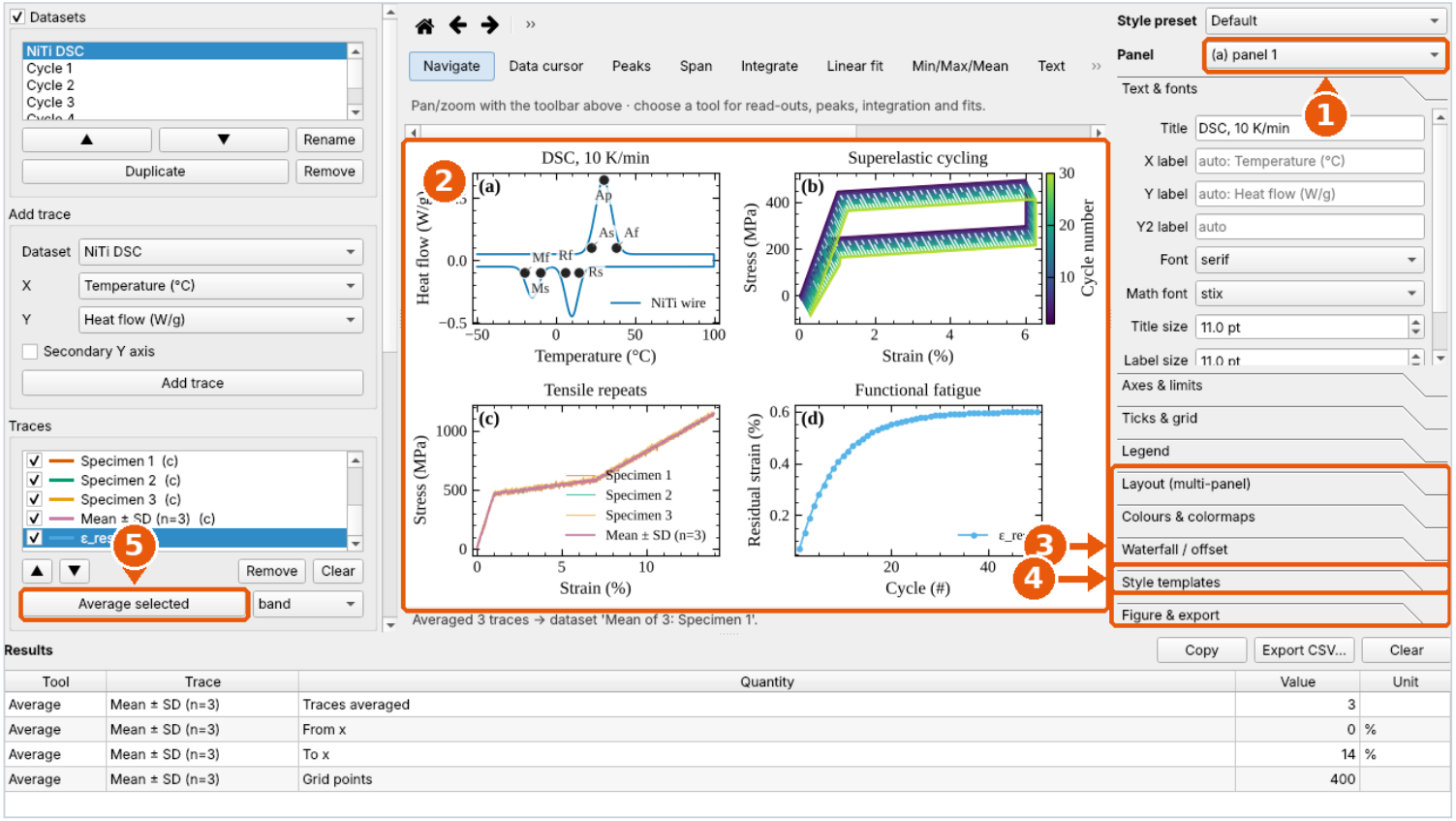
2 Interactive tools
Navigate · Data cursor · Peaks · Span · Integrate · Linear fit · Min/Max/Mean · Text. See the next page.

3 Datasets & trace style
Top-left list: rename, reorder, duplicate or remove datasets. Select a trace to change label, colour, line, marker, panel and error band.

4 Customise panel
Title, labels, limits, log scale, display units (GPa, K…), fonts, ticks, grid, legend, multi-panel layout, palettes, offsets, style templates and export, all live.

5 Results table
Every tool writes here. Amber rows are warnings. Copy uses your significant-figures setting; Export CSV adds parameters, source file and checksum at the top.

Figure styling: templates, panels, palettes



1 Panel selector
With a multi-panel layout, choose which panel the title, labels and limits apply to.

2 Multi-panel figure
Layout 1×2, 2×2, 3×1… with automatic (a), (b), (c) labels and shared axes. Assign each trace to a panel in its style box.

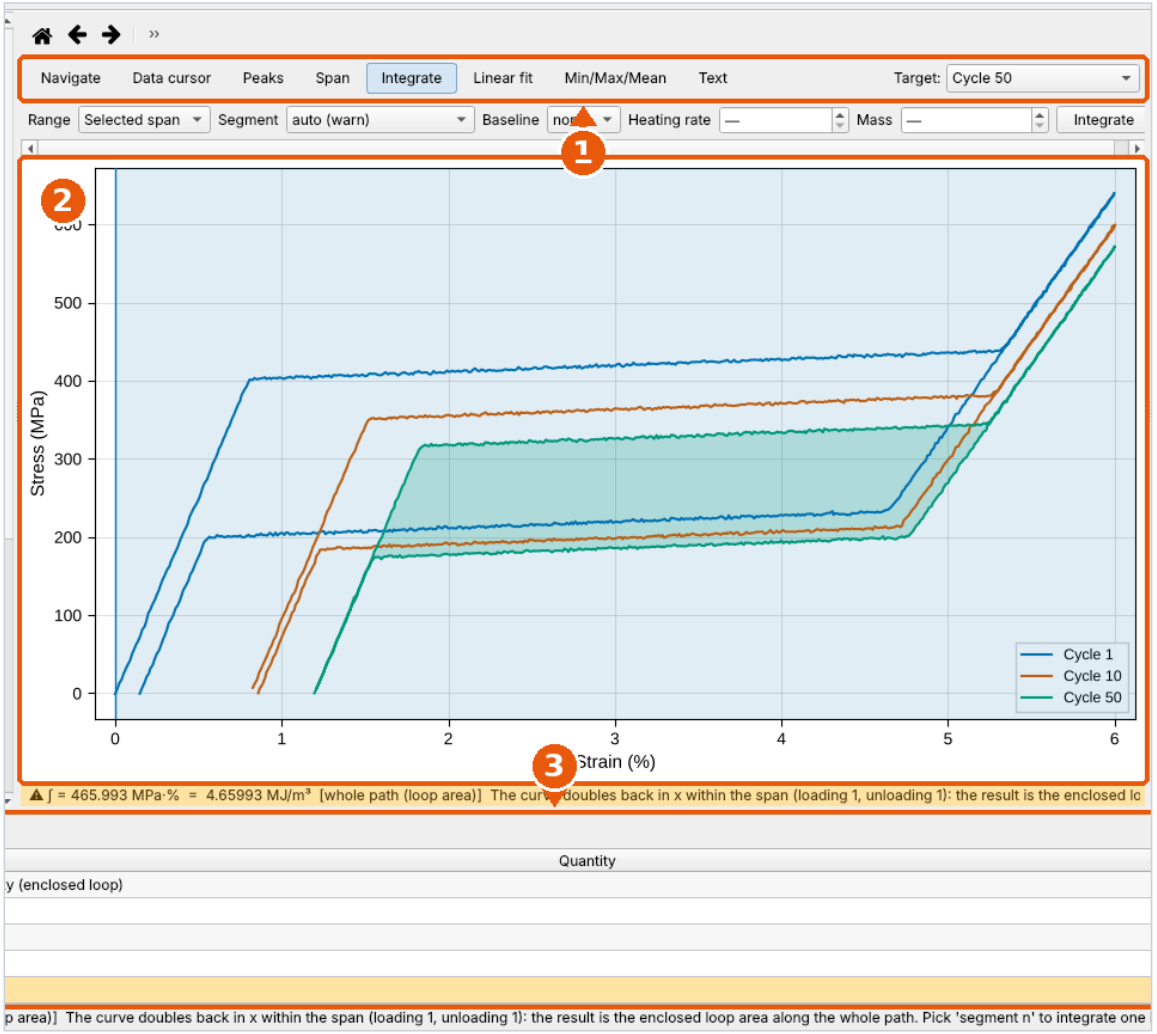
3 Layout · Colours · Waterfall
Colours & colormaps: Okabe–Ito, Tableau, greys or viridis steps, or colour traces by a metadata value with a colorbar. **Waterfall / offset** stacks DSC scans (display only).

4 Style templates
Built-in Thesis (Imperial, serif 11 pt, 15 cm), Journal single column, Slides, or save your own. Apply to all new figures keeps every chapter identical.

5 Average selected
Select repeat specimens → mean ± SD band (or error bars) as a new dataset, n recorded.

Preview at export size (Figure & export) shows the figure at its exported cm size, so labels never shift between screen and thesis.

Reading values, peaks and integrating



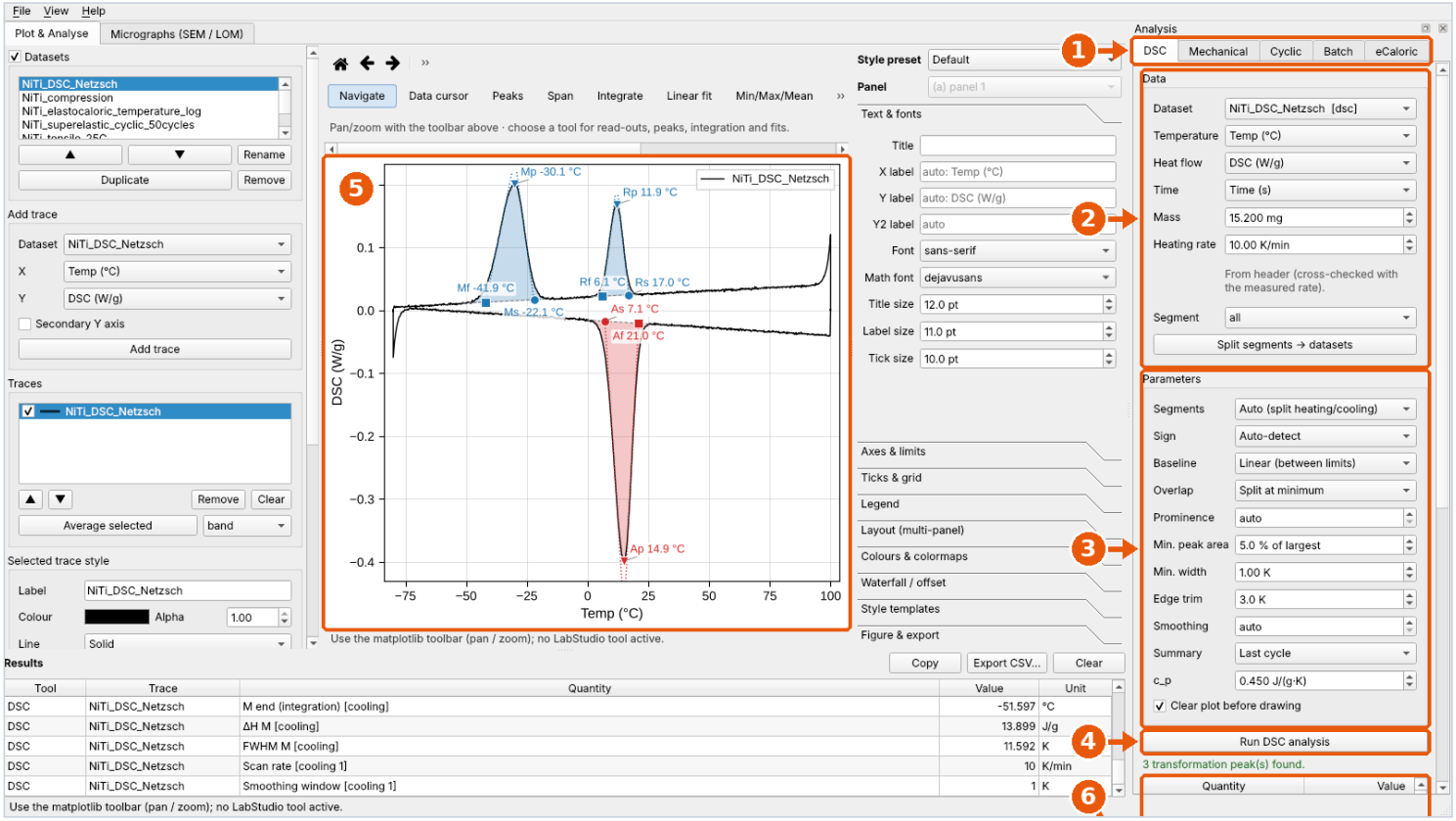
Tool	What it does	How
Data cursor	Read x, y of the nearest data point (snaps across all traces)	Hover; click to pin → results
Peaks	Find peaks or valleys with prominence, position and FWHM	Choose trace, set prominence, Find
Span	Select an x-range used by the other tools	Drag across the plot
Integrate	Area of the curve or span, optional linear baseline; Segment = auto / loop / heating n	Range + baseline, Integrate
Linear fit	Slope, intercept, R^2 on the span (E in GPa for stress–strain)	Select span, Fit
Min/Max/Mean	Statistics of y on the span	Select span
Text	Add draggable labels to the figure	Click where the text goes

Integration gives physical units

- stress (MPa) × strain (%) → **MJ/m³**: the closed loop area is the dissipated energy per cycle (shown in the screenshot: 4.66 MJ/m³)
- force (N) × displacement (mm) → **J**
- heat flow (mW or W/g) vs time → **J** or **J/g**
- heat flow vs temperature → divided by the heating rate (pre-filled from the file) → **J/g**
- a DSC trace with heating and cooling is integrated one segment at a time, with a warning

Numbered callouts: (1) the tool bar, (2) drag on the plot to choose the span (shaded green = integrated area), (3) the result appears in the table with its unit.

DSC: transformation temperatures and latent heat



1 Analysis dock
Tabs DSC | Tensile / Compression | Cyclic, docked beside the plot (View menu to show or hide).

2 Data
Dataset and columns; mass and heating rate from the file (source shown). **Segment**: all, heating n, cooling n, last heating… **Split segments → datasets** for clean overlays.

3 Parameters
Sign convention, baseline, **overlap** (split at minimum or combined R+M), minimum peak area and width (rejects noise), smoothing, c_p. Peak limits can be edited by hand.

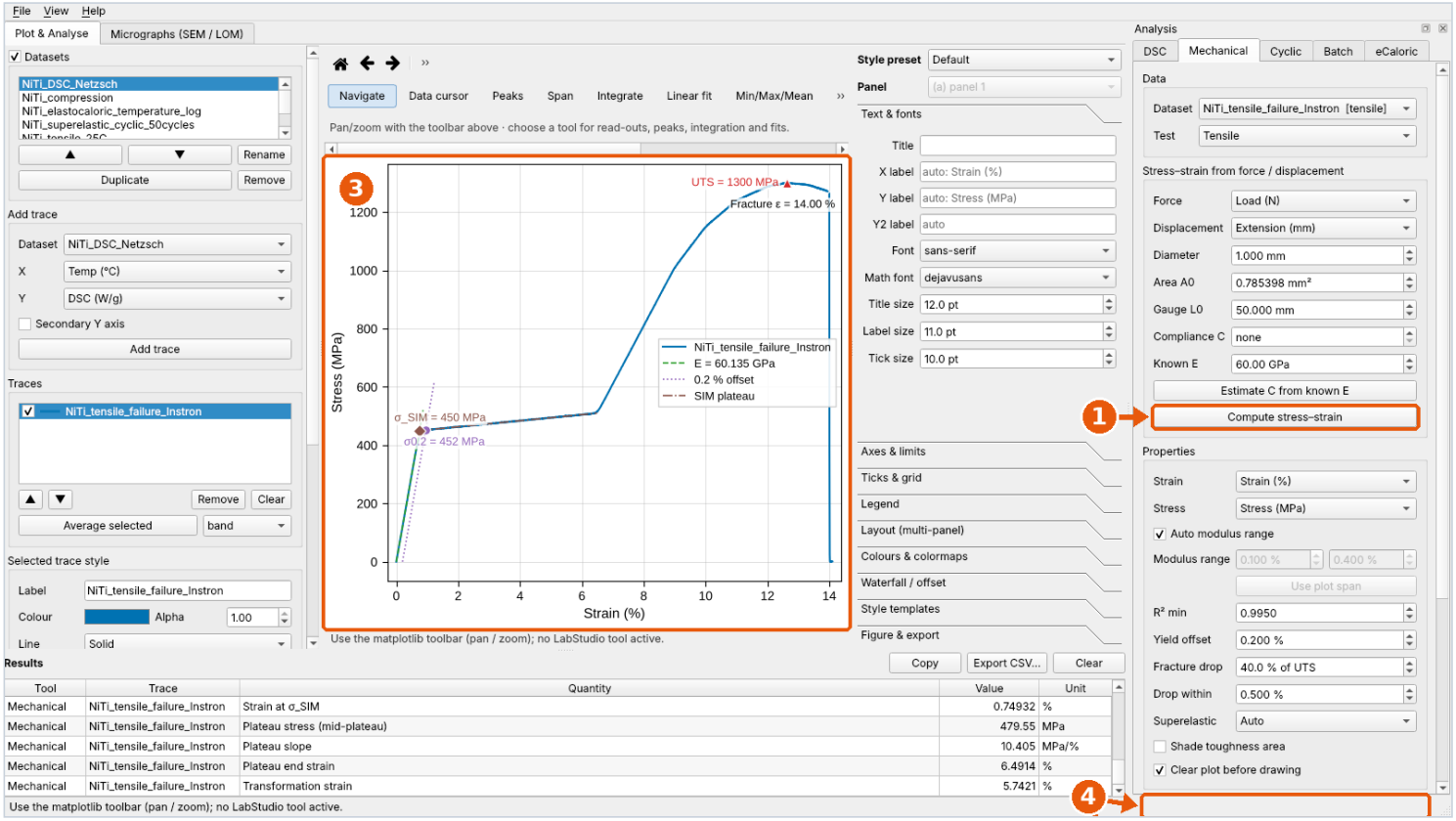
4 Run DSC analysis
Finds each peak and builds the tangent onsets.

5 On the plot
As, Ap, Af (heating) and Rs, Rf, Ms, Mp, Mf (cooling) marked, with the integrated areas shaded.

6 Results
Temperatures (°C), ΔH (J/g), hysteresis, ΔT_{ad} upper bound. Amber warnings flag rate mismatches, truncated peaks and heating/cooling ΔH differing by more than 20 %.

Method: onset / end = intersection of the baseline with the tangent at the steepest point of the peak edge (standard tangent method); $\Delta H = \int(q - \text{baseline})\,dt \div \text{mass}$.

Tensile and compression tests



1 Compute stress–strain
Converts force and displacement into engineering stress $\sigma = F/A_0$ (MPa) and strain $\varepsilon = \Delta L/L_0$ (%) using the specimen geometry, then plots σ–ε.

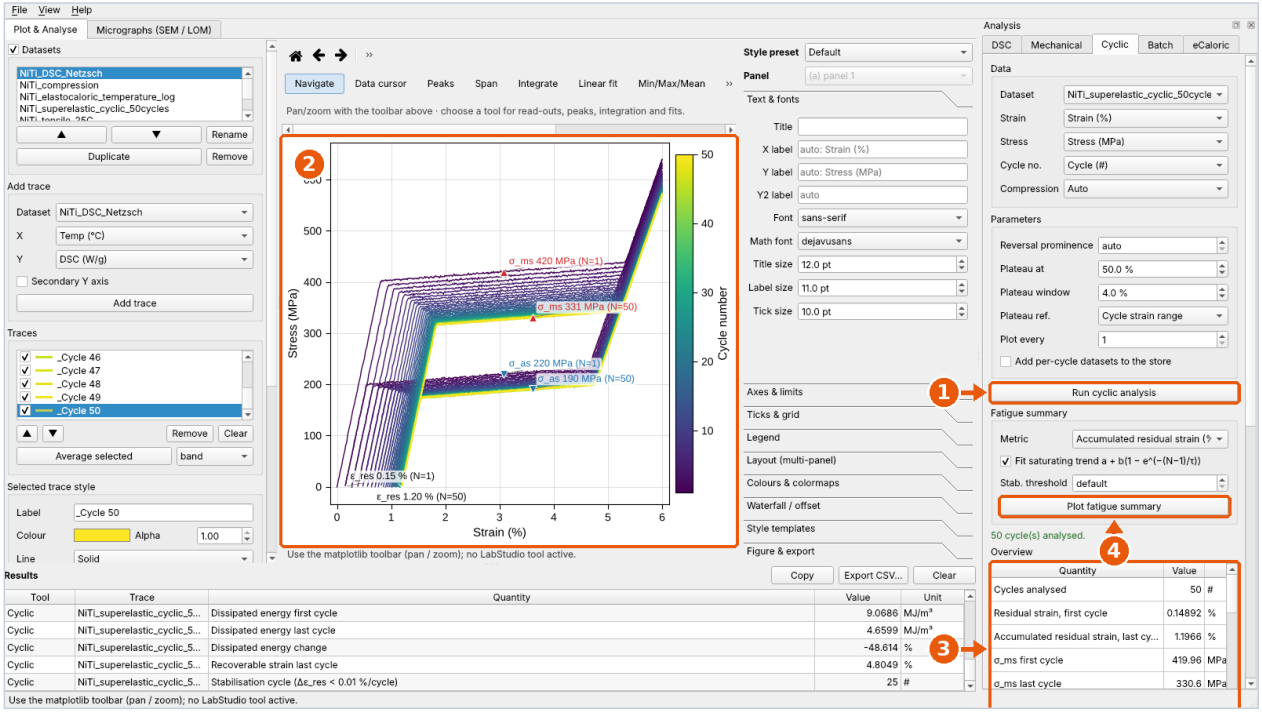
2 Run mechanical analysis
Modulus range automatic or from a plot span. Optional machine compliance C (or Estimate C from known E) corrects crosshead strain.

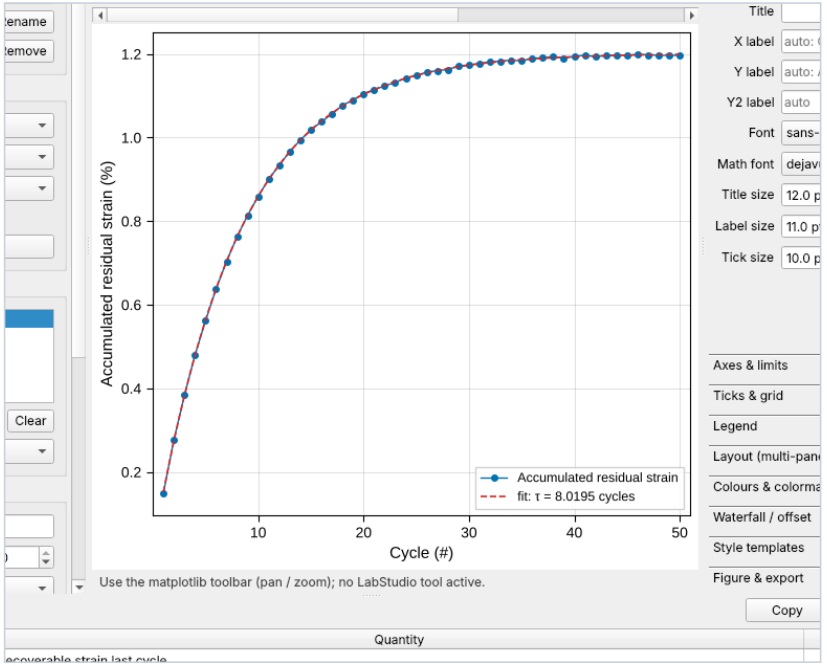
3 Overlays
Modulus line, 0.2 % offset line, σ_{SIM} (start of the superelastic plateau), UTS and fracture point.

4 Results
E (GPa), 0.2 % yield, UTS, fracture strain, toughness (MJ/m³), σ_{SIM}, plateau, transformation strain. Impossible values (E > 300 GPa…) are flagged.

Compression: load a compression file and set Test type = Compression; the sign convention is handled automatically (negative machine values become positive stresses).

Cyclic superelastic tests and functional fatigue





Result of button 4: accumulated residual strain vs cycle number

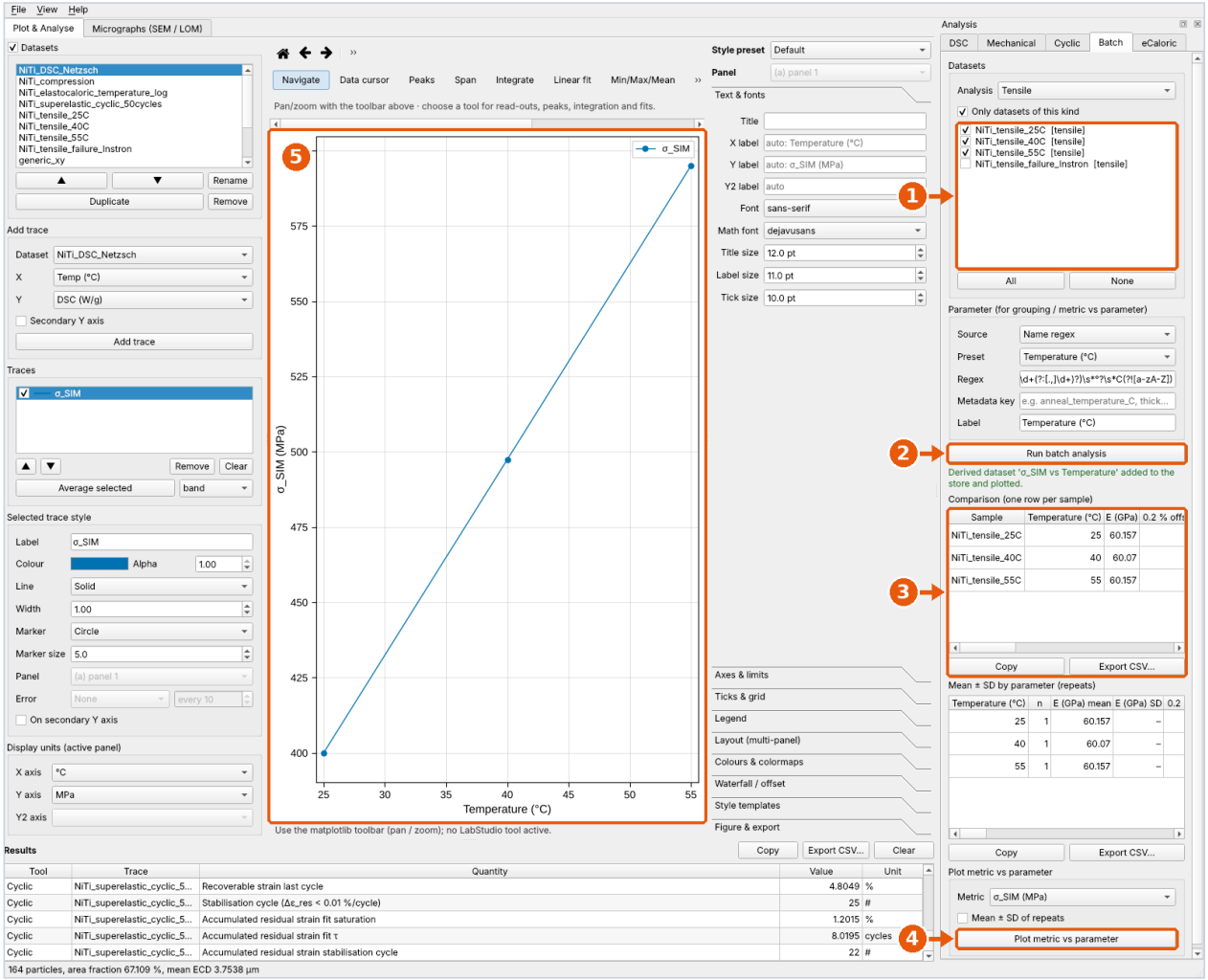
1 Run cyclic analysis
Splits cycles (column or load reversals), colours them with a cycle-number colorbar. Compression data are sign-corrected automatically; force/displacement files get their own stress–strain step.

2 On the plot
σ_{ms} (upper / forward plateau) and σ_{as} (lower / reverse plateau) read at mid-plateau strain, residual strain ε_{res} marked on the first and last cycle.

3 Per-cycle table
Max stress and strain, residual and recoverable strain, dissipated energy (loop area, MJ/m³), efficiency, σ_{ms}, σ_{as}, $\Delta\sigma$.

4 Plot fatigue summary
Metric vs cycle number plus a saturating trend fit: τ, saturation value and stabilisation cycle. Incomplete cycles are flagged and excluded.

Batch analysis: many samples, one table



1 Pick datasets
Batch tab: choose the analysis (DSC, tensile, compression, cyclic) and tick the samples. The parameters of that analysis tab are used for all of them.

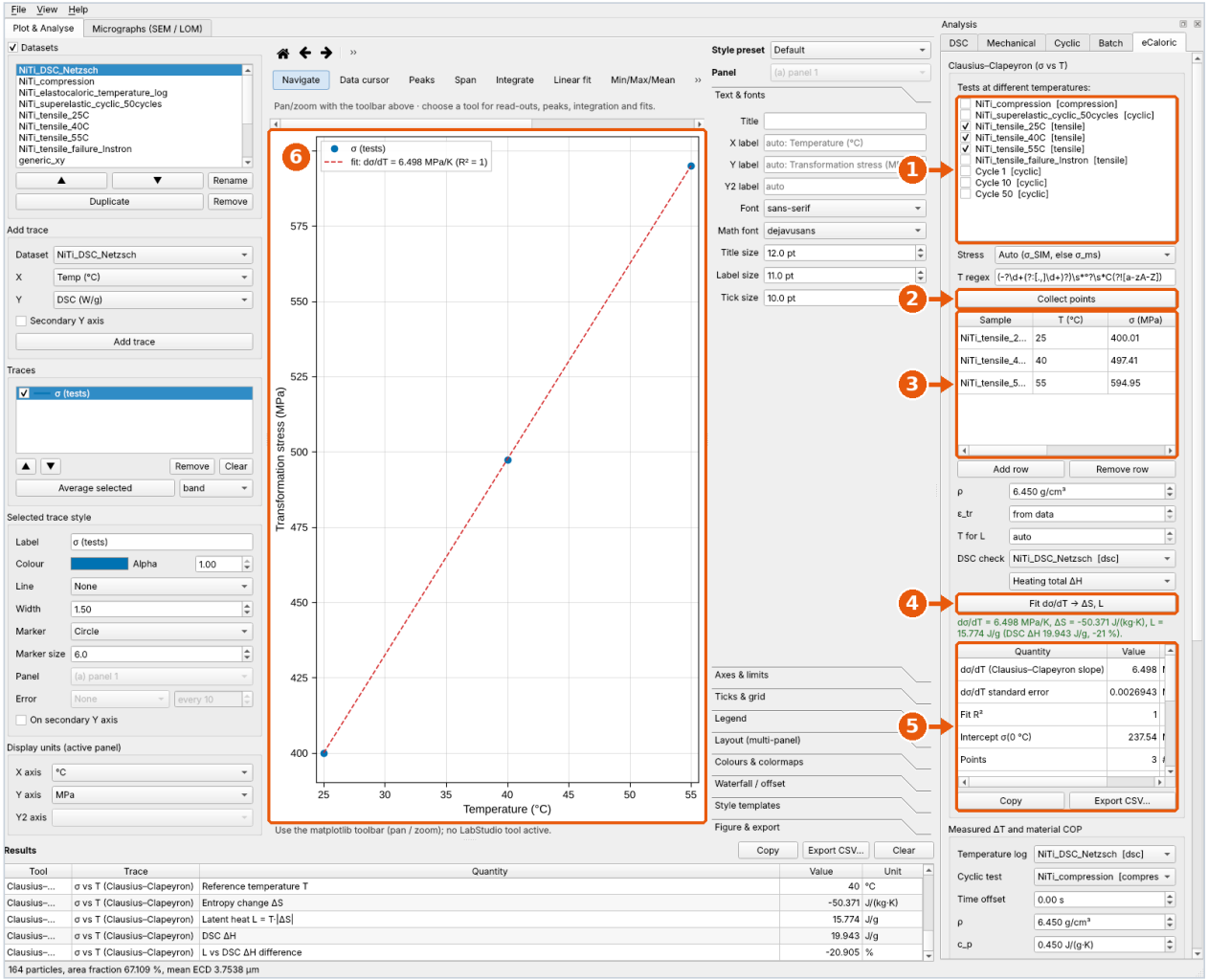
2 Run batch analysis
One row per sample with the key results and a warnings column; errors are reported per row instead of stopping the run.

3 Comparison table
Export CSV (with provenance). The parameter (temperature, strain, thickness…) is read from the file name with a regex preset or from a metadata key.

4 Plot metric vs parameter
e.g. σ_{SIM} vs test temperature; tick Mean ± SD of repeats to group repeats. Creates a dataset you can style like any other.

5 Result
The derived dataset is plotted straight away.

Elastocaloric: Clausius–Clapeyron, ΔS and L



1 Tests at different T
eCaloric tab: tick tensile or cyclic tests run at different temperatures (temperature from the file header or name).

2 Collect points
Fills the table with T, σ_{SIM} (or σ_{ms}) and ε_{tr}; rows can be edited or added by hand.

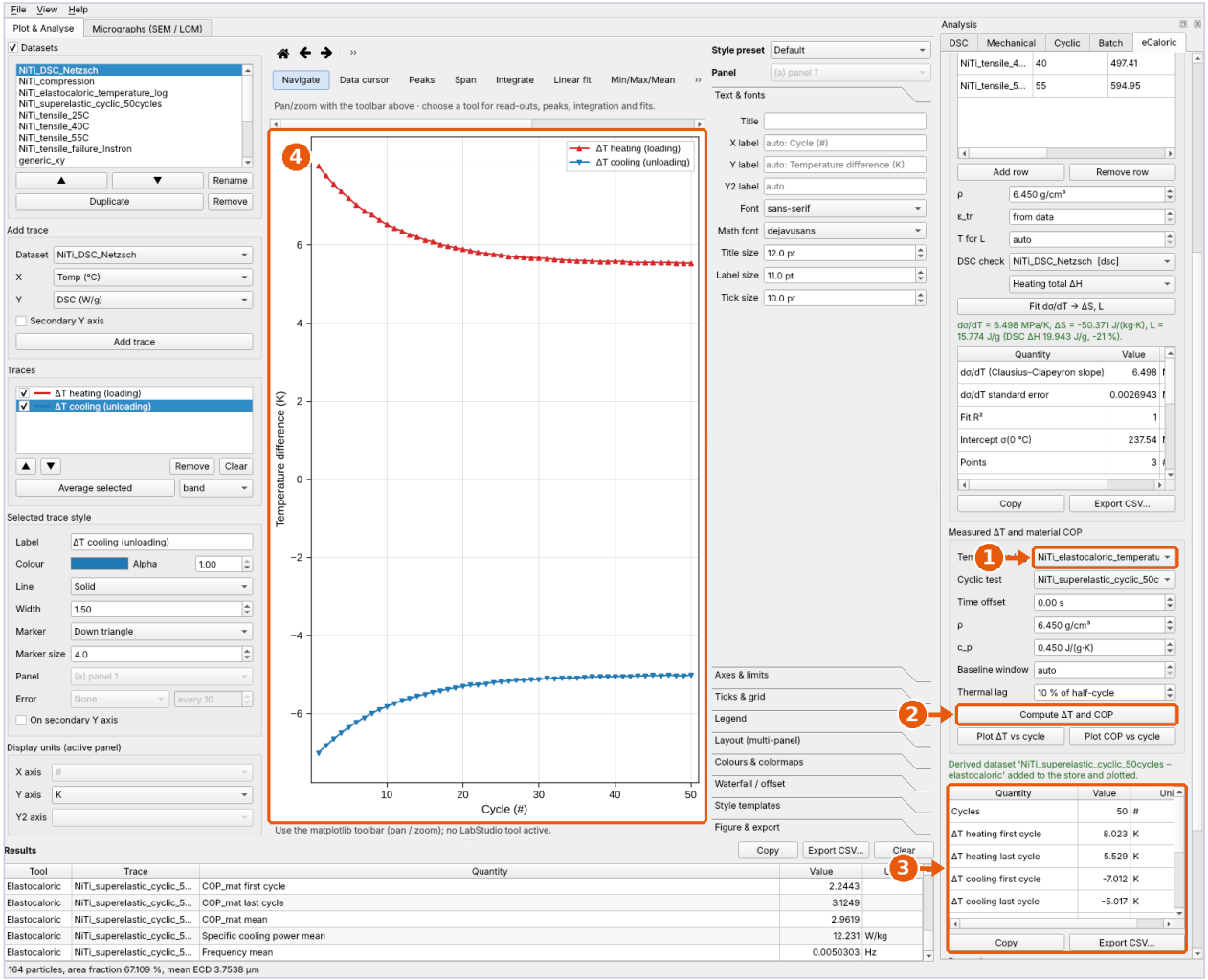
3 Points table
Check the values before fitting; set ρ and ε_{tr} below if needed.

4 Fit dσ/dT → ΔS, L
Linear fit with R^2, $\Delta S = -(d\sigma/dT)\cdot\varepsilon_{tr}/\rho$ in J/(kg·K) and $L = T\cdot|\Delta S|$ in J/g.

5 DSC cross-check
Pick a DSC dataset to compare L with the measured ΔH (difference in %).

6 Plot
σ vs T with the fitted line (sample data: 6.50 MPa/K).

Elastocaloric: measured ΔT, COP and cooling power



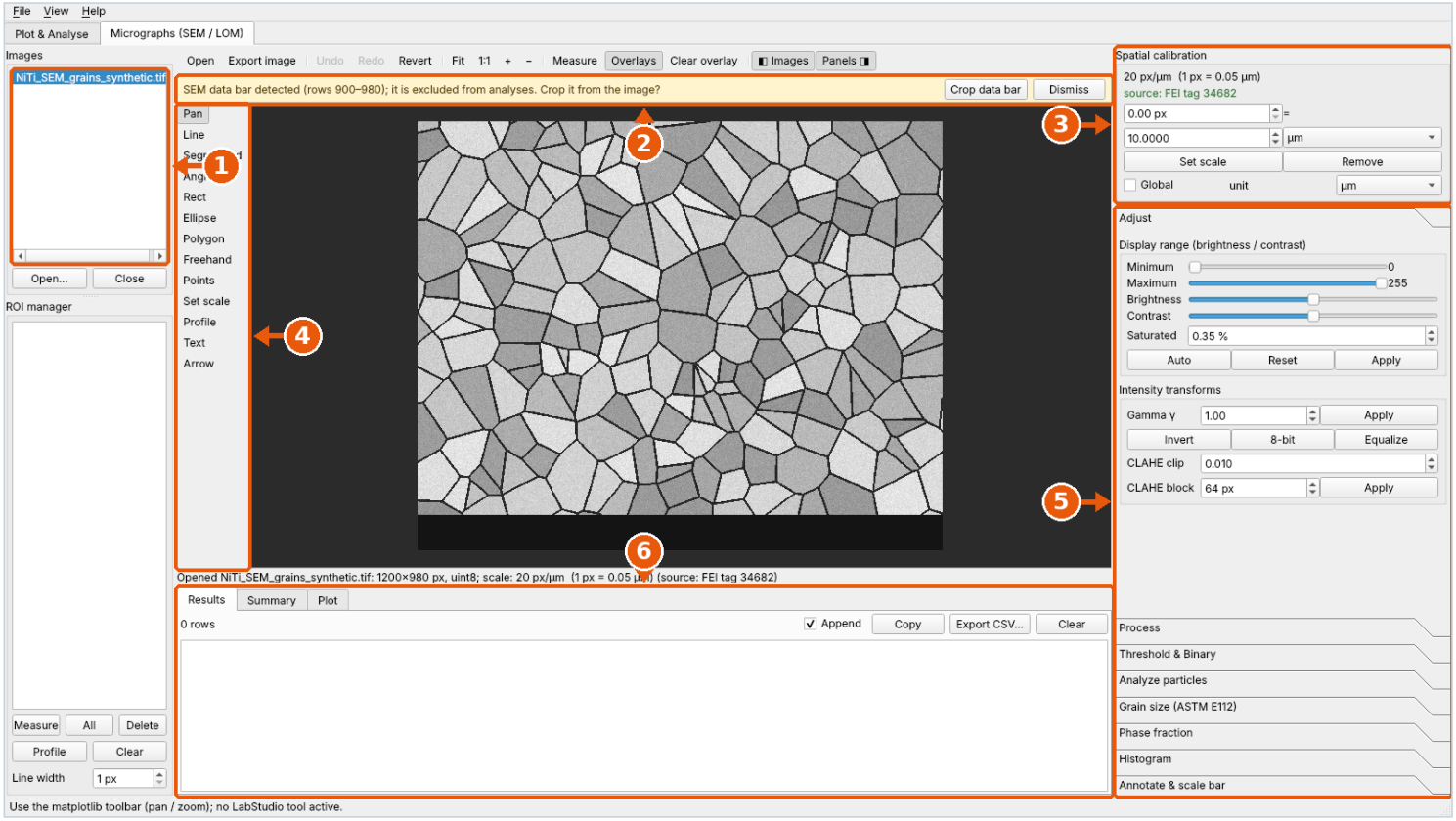
1 Temperature log + cyclic test
Choose the thermocouple / IR log and the cyclic test; set a time offset if the clocks differ, plus ρ, c_p and the baseline window.

2 Compute ΔT and COP
Per cycle: ΔT on loading and unloading vs the pre-cycle baseline, input work W (loop area), $Q_c = c_p \cdot |\Delta T_{cool}|$, $COP_{mat} = Q_c / W$, specific cooling power (W/kg).

3 Summary
First / last / mean COP, cooling power and frequency; the full per-cycle table exports to CSV.

4 Plot ΔT / COP vs cycle
Derived datasets showing how the caloric effect fatigues.

Opening and calibrating SEM / optical images



1 Open images
TIFF (8 / 16-bit, multi-page), PNG, JPG, BMP via Open… or drag & drop. Several images can be open at once.

2 Crop the SEM data bar
Detected automatically; one click removes it so it isn't measured.

3 Spatial calibration
From FEI, Zeiss, JEOL/Hitachi or ImageJ metadata, with the **source shown**. Print-dpi images open uncalibrated: **Set scale** along the scale bar instead.

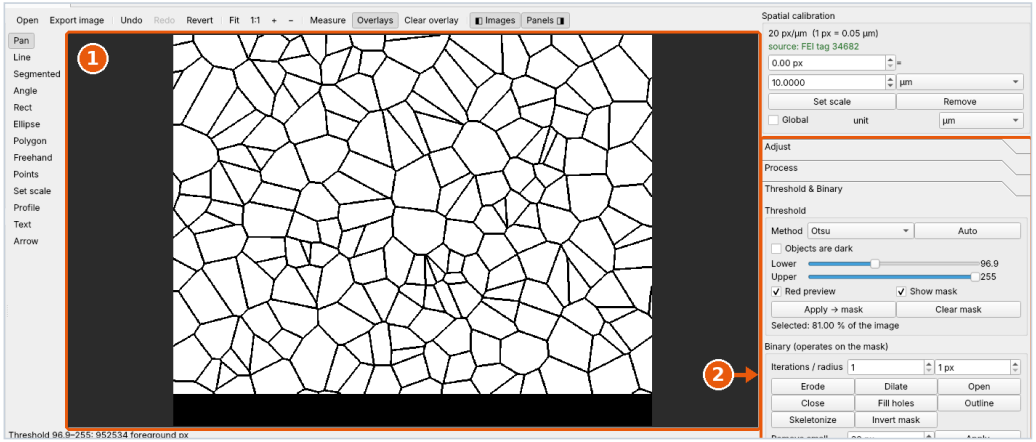
4 Tools
Line, segmented line, angle, rectangle, ellipse, polygon, freehand, points (counter), profile, text, arrow. Press **Measure** to send an ROI to the results.

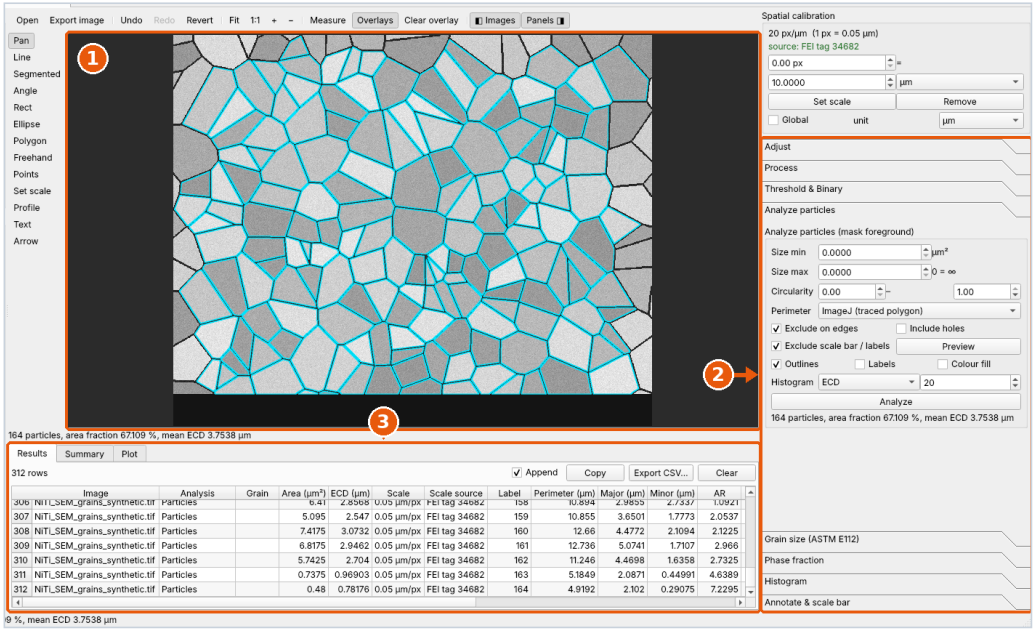
5 Panels
Adjust · Process · Threshold & Binary · Analyze particles · Grain size (ASTM E112) · Phase fraction · Histogram · Annotate & scale bar.

6 Results / Summary / Plot
ImageJ-style tables in calibrated units, plus histograms. Copy or Export CSV.

> **Adjust** covers brightness / contrast, gamma, CLAHE, invert and 8-bit; **Process** covers filters, FFT band-pass, rolling-ball background, crop / rotate / resize. Undo / Redo / Revert at the top.

Threshold, binary and particle analysis





Threshold (top image)

1 Binary mask
While you move the sliders the selected pixels are previewed in red; after Apply the binary mask (shown here) is what all binary operations and measurements use.

2 Threshold & Binary panel
Manual sliders or auto: Otsu, Li, Yen, Triangle, IsoData, Mean, Minimum, MaxEntropy; dark or light objects. Then binary operations: erode, dilate, open, close, fill holes, watershed (separate touching particles), skeletonize.

Analyze particles (bottom image)

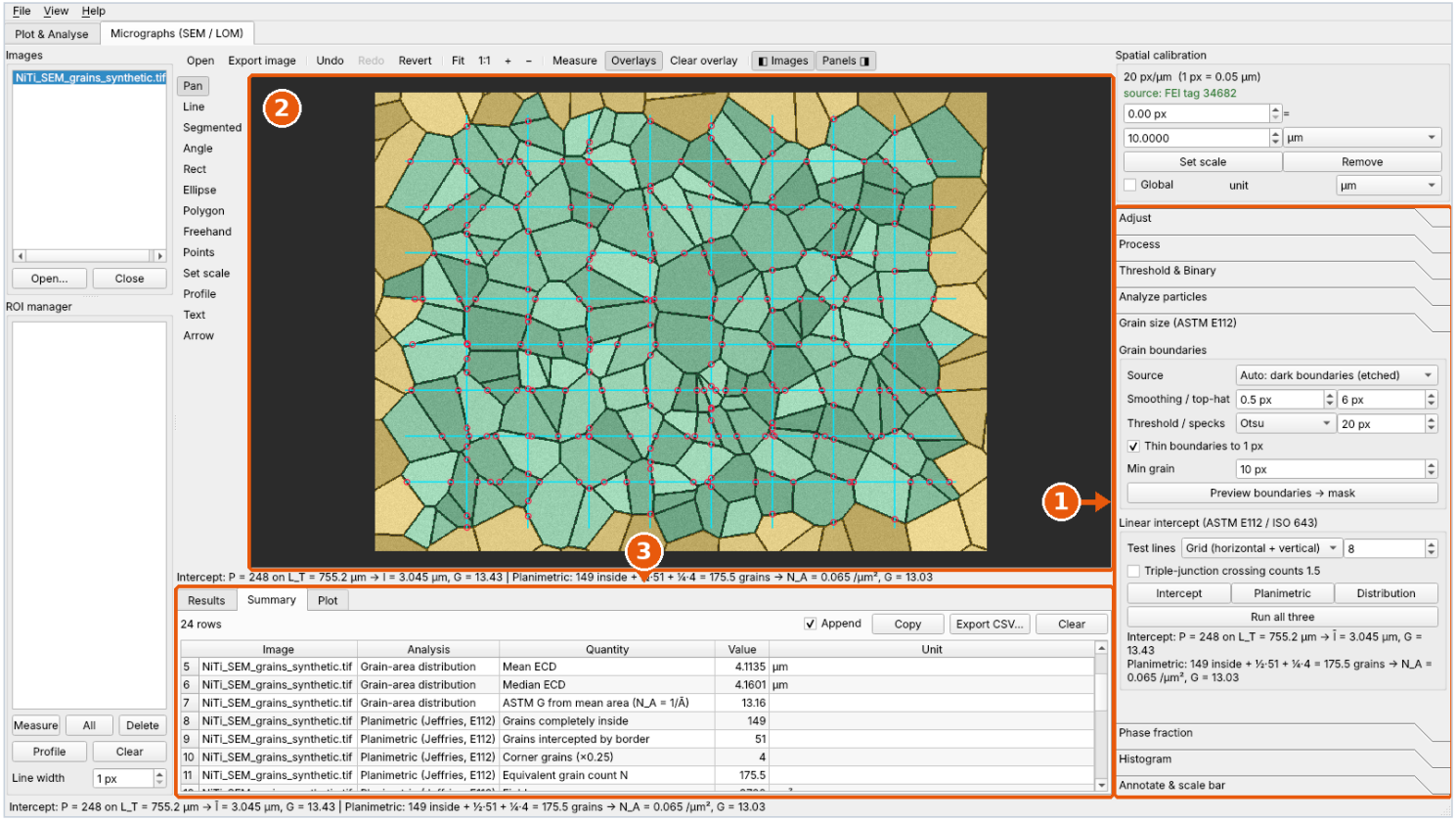
numbering restarts on this image

1 Outlines
Every detected particle is outlined (or labelled / colour-filled).

2 Settings
Size and circularity range, exclude edges, **exclude scale bar / labels** (preview), perimeter method (ImageJ traced polygon by default, so values match ImageJ).

3 Per-particle table
Area, ECD, perimeter, major / minor, aspect ratio, circularity $4\pi A/P^2$, roundness, solidity, Feret max / min / angle, centroid. Summary tab: count, area fraction, mean size.

Grain size (ASTM E112) and phase fraction



1 **Grain size panel**
Boundaries: auto from the etched image (dark or bright) or from your threshold mask. Pattern: horizontal / vertical / grid lines or three circles. **Run all three** = intercept + planimetric + distribution.

2 **Overlay**
Test lines (cyan) with the counted boundary intersections; planimetric colouring (green = inside grains, yellow = edge grains).

3 **Results**
Mean lineal intercept $\bar{l}$ (µm) and $G = -3.2877 - 6.6439 \log_{10}(\bar{l}\ \text{mm})$; Jeffries N_A (grains/mm²) and $G = 3.3219 \log_{10} N_A - 2.954$; grain-area / ECD distribution.

Phase fraction: multi-level Otsu or manual thresholds. **Annotate & scale bar**: presets (Thesis / Journal / Minimal or your own), apply to all images, µ renders correctly. Fewer than 50 intercepts or grains → low-statistics warning.

Sessions, warnings and preferences

Save your work as a session

File ▸ Save session (Ctrl+S) writes one **.lstudio** file with every dataset, figure, trace style, analysis setting and open micrograph (with its calibration, ROIs and results). Open it later with **File ▸ Open session**, **Open recent** or by dropping it on the window. The title shows • when there are unsaved changes.

No silent wrong answers

Whenever a result depends on a guess or looks suspicious, Materia says so in **amber**: guessed units, default heating rate, header vs measured rate, truncated peaks, heating/cooling ΔH mismatch, incomplete cycles, impossible moduli, print-dpi calibration, low grain statistics. Warnings travel with the data into every CSV.

Traceable numbers

Every exported CSV starts with **provenance** lines: Materia version, date, analysis, all parameters (including automatic ones), source file name and its SHA-256 checksum. **File ▸ Preferences** sets significant figures for tables and copy, full-precision CSV and the default figure template. **Help ▸ Report a problem** saves the log for bug reports.

Small screen? Both tabs fit a 1280×720 laptop: side panels scroll and can be hidden from the toolbar, and **View ▸ Fit window to screen** resizes the window. Long jobs (large cyclic files, 8k micrographs) run in the background with a busy indicator.

Quick reference and good practice

Units used throughout

Quantity	Unit
Force	N
Displacement, length	mm
Area	mm²
Stress	MPa
Strain	%
Temperature	°C
Time	s
Heat flow	mW or W/g
Enthalpy	J/g
Energy density	MJ/m³
Modulus	GPa

Which tab for which test

Data	Where
DSC scan	Analysis ▸ DSC
Tensile to failure	Analysis ▸ Tensile
Compression	Analysis ▸ Tensile / Compression
Superelastic cycling	Analysis ▸ Cyclic
Many samples	Analysis ▸ Batch
σ vs T, ΔT, COP	Analysis ▸ eCaloric
Anything x–y	Plot + tools
SEM / LOM image	Micrographs tab

Good practice for the dissertation

• Save a **preset** for each machine on day one; consistent imports mean consistent results.
• **Save a session** per chapter or sample batch, and never ignore an amber warning.
• Always check the **specimen geometry** in the import dialog: stress is only as good as A_0 and L_0.
• Use the **Publication** style preset and export at **300–600 dpi** for figures.
• Export the **results tables as CSV** next to each figure so every number is traceable.
• For DSC, analyse the **second** (or last) thermal cycle; the first can carry processing history.
• Set the SEM scale from metadata or the scale bar **before** measuring anything.
• Report your methods: tangent onset for DSC, 0.2 % offset yield, ASTM E112 intercept for grain size.

File ▸ Load sample NiTi data reloads the demo files at any time. Shortcuts: Ctrl+S save session · Ctrl+O import · Ctrl+Shift+O image · Ctrl+E export · Ctrl+, preferences.